Module 6: PRS Portability on Real Data – Alzheimer's Disease

1. Objective and data provenance

This analysis evaluated how a European-derived polygenic risk score (PRS) transfers across ancestry groups in a 1000 Genomes genotype panel. The score was aligned with genotype alleles and calculated for each individual. We then evaluated its distribution across ancestry groups, predictive performance, and added value beyond age. Genotypes and score inputs are real; disease case/control labels are simulated. Accordingly, all reported AUCs and calibration curves are illustrative, not validated estimates of clinical Alzheimer's disease prediction.

```
source      : REAL
people      : 2336
score size  : 22 variants
by ancestry : {'AFR': np.int64(500), 'EAS': np.int64(500), 'EUR': np.int64(500), 'SAS': np.int64(489), 'AMR': np.int64(347)}
  genotypes.csv: sha256 9cf41a1f
  variants.csv: sha256 f7a87d3f
  score.csv: sha256 7f80e8ff
  meta.csv: sha256 6cc55b6e
```

Figure 1. Proof of data (provided output of Section 2)

2. Allele harmonization and PRS calculation

Effect alleles were compared with the genotype panel's ALT and REF alleles. ALT-matched variants kept their original dosages. REF-matched variants were flipped (2 − dosage) to count the effect allele. Variants matching neither allele would have been excluded. All 22 score variants were present in the genotype panel: 22 were used, 13 required flipping, none was dropped for allele mismatch, and none was absent. Individual PRS values were calculated as the sum of aligned effect-allele dosages multiplied by their corresponding weights, then standardized across the full cohort.

3. Ancestry distribution and threshold fairness

The whole-cohort 90th-percentile PRS cutoff was 1.493. The proportion above this single cutoff varied markedly by ancestry, despite the nominal "top 10%" label. 28.0% of AFR participants exceeded it compared with only 2.2% of EAS participants. AFR also had the highest mean standardized PRS (0.744), while EAS had the lowest (−0.472). These shifts reflect the non-equivalence of a pooled PRS threshold across ancestry groups; they cannot by themselves establish differences in true disease incidence or genetic susceptibility.

Ancestry (n)	Mean PRS	Above 1.493	AUC
EUR (500)	0.036	7.8%	0.672
AFR (500)	0.744	28.0%	0.493
EAS (500)	−0.472	2.2%	0.575
SAS (489)	−0.245	4.3%	0.611

AMR (347)	−0.099	6.3%	0.555

If a clinic used this pooled cutoff unchanged, it would flag substantially different fractions of people from different ancestry groups. In these data, the AFR group would be flagged much more often even though its score has almost no case/control discrimination. This risks unequal follow-up burdens and misleading risk communication. Group-specific cutoffs could equalize selection rates, but they would not necessarily improve discrimination or calibration. Clinical thresholds should be validated using observed outcomes while considering potential benefits and harms.

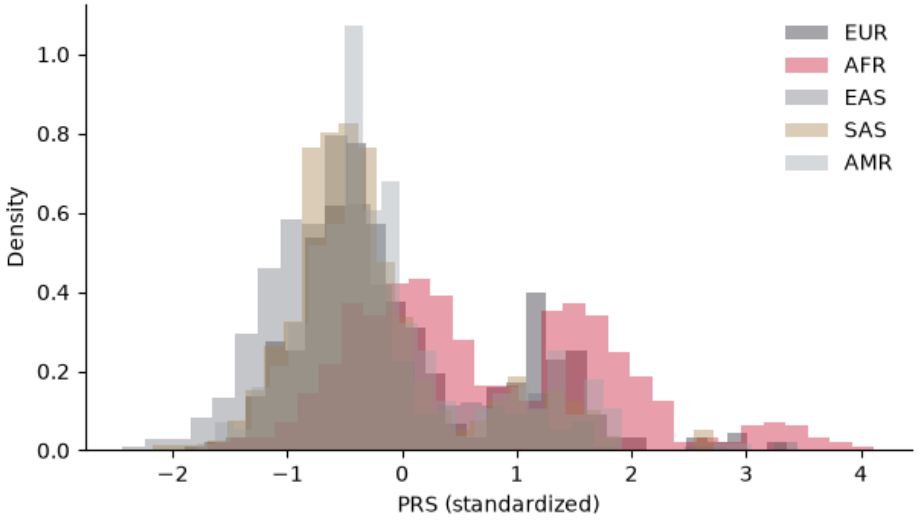


Figure 2. Standardized PRS distribution by ancestry

4. Discrimination versus calibration

The overall AUC was 0.599, indicating modest discrimination in the simulated case/control data. EUR had the highest AUC (0.672), followed by SAS (0.611), EAS (0.575), AMR (0.555), and AFR (0.493). An AUC near 0.5 means the score provides essentially no ranking information for simulated cases versus controls in that group. Importantly, the high mean PRS and high threshold-exceedance fraction in AFR coexist with an AUC of 0.493; a shifted score distribution is not evidence of useful within-group discrimination.

Discrimination asks whether cases tend to rank above controls and calibration asks whether a predicted probability agrees with the observed case frequency. The logistic risk model was fitted only in EUR and applied unchanged to all groups. In the calibration plot, points deviate from the ideal diagonal in multiple groups, including EUR. The AFR curve is visibly irregular and includes substantial discrepancies between predicted and observed proportions. Therefore, a model can produce high scores or even moderate ranking performance without delivering reliable absolute risk estimates. The plotted curves cannot support a precise ranking of calibration error without additional quantitative metrics and uncertainty intervals.

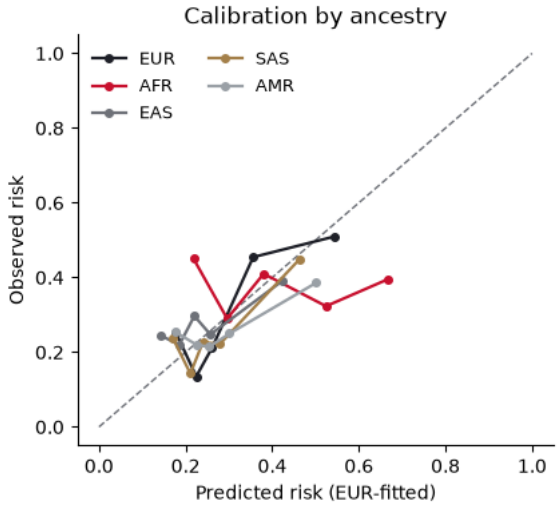


Figure 3. Calibration by ancestry using a logistic model fitted in EUR

5. Why portability declines

A PRS developed primarily in European-ancestry datasets may lose predictive performance elsewhere because linkage disequilibrium (LD) patterns and allele frequencies differ across populations. A score variant that tags a causal allele well in EUR may be a weaker proxy in another ancestry, even if the causal biology is shared. Differences in score-variant frequency can also shift the aggregate PRS distribution. Statistical differences and training-data imbalance may explain why PRS performance varies across populations. This does not necessarily mean that the underlying disease biology differs. The present simulation illustrates this mechanism but cannot identify the unique cause of every observed difference.

6. Incremental value, clinical utility, and limits

Five-fold cross-validated AUC increased from 0.561 for age alone to 0.618 for age plus PRS, an absolute gain of 0.057. The score modestly improved discrimination in the simulated cohort, but overall performance remained limited. This improvement may not apply equally across ancestry groups. The score may be useful for research on PRS transferability or for developing future risk models. However, it is not suitable for clinical screening, diagnosis, or treatment decisions without further validation.

Before clinical use, the score must be validated in independent cohorts with real Alzheimer's disease outcomes and accurate clinical data. These cohorts should include direct APOE genotypes and sufficient representation of each ancestry group. External validation should assess ancestry-specific AUC, calibration, and uncertainty. It should also examine the clinical benefits and risks of different thresholds. Prospective studies should address informed consent,

privacy, and potential disparities in clinical outcomes. Simulated phenotype labels alone cannot establish clinical accuracy or utility.

7. Artifact versus biology: APOE ε4

The near-chance AFR AUC (0.493) must not be interpreted as evidence that APOE ε4 is unimportant in people of African ancestry. The assignment specifies that the simulated dataset contains an LD proxy for APOE ε4 that is almost absent in the AFR group. As a result, the chosen proxy contributes little useful signal for the simulated phenotype in that group. This reflects how the dataset was constructed and how proxy variants were selected, not a biological testing of APOE ε4 itself. Direct APOE ε4 genotyping and independent real-world outcome data would be necessary to evaluate its contribution across ancestry backgrounds. Proxy failure should not be confused with absence of causal effect.